A Critical-Depth Row-Mass Obstruction for Dilated Prime-Shell Clocks

Liang Wang
Huazhong University of Science and Technology

August 24, 2026

Abstract

We audit a hidden conditioning hypothesis in a growing-depth prime-shell resonance model. At depth L, modulus $4LQ$, and prime row $Q < q < 2Q$, the primitive support contains the signed multipliers $|m| \leq \lfloor Lq/Q \rfloor$. Previous incidence bounds transfer to energy only when row masses have a scale-independent comparability constant. We construct a critical sequence $L \sim \log Q / \log\log Q$ for which this constant diverges. Taking Q to be a power of two times the product of all primes at most L, a low-shell prime row has exactly two atoms, whereas a high-shell row has $2\{1 + \pi(2L-1) - \pi(L)\}$ atoms. Thus the raw mass ratio is at least $(1+o(1))L/\log L$. A universal upper bound $2L-1$ shows the size of the conditioning window. Four exact finite clocks are reproduced by independent deterministic primality implementations. The result refutes automatic raw comparability, but does not rule out row normalization or identify the model with the physical arithmetic source.

1 Introduction

Sparse collision geometry can obstruct a fixed proportional energy saving. Such a support statement becomes a mass statement only after controlling how much diagonal energy a single row may carry. In the dilated-clock model considered here, that transfer was previously isolated as a fixed row-mass comparability hypothesis. The purpose of this paper is to determine whether the clock itself supplies that hypothesis at the first depth not already excluded by a sieve upper bound.

The answer is no. We exhibit a sequence at the critical scale $L \asymp \log Q / \log\log Q$ on which one prime row has only the two atoms generated by $m = \pm 1$, while another row has one signed pair for every prime multiplier in $(L, 2L)$. The resulting mass ratio tends to infinity. This is an obstruction to a particular transfer assumption, not to the resonance model as a whole.

The construction combines two elementary mechanisms. Primorial saturation makes every integer $2 \leq m \leq L$ nonprimitive. A classical prime number theorem remainder places active prime rows in short intervals at the two ends of $(Q, 2Q)$. We use the standard zero-free-region form of the PNT; a bare asymptotic formula for $\pi(x)$ would not justify subtraction over these shrinking relative intervals [1, 3].

The main contribution is therefore an exact conditioning counterexample with an arithmetic shell-placement theorem. It leaves two deliberate boundaries. First, normalizing each row can remove raw mass imbalance, and its collision operator must be analyzed separately. Second, uniform atom weights are part of this modeled clock and are not the actual source coefficients.

2 Raw row mass and a universal envelope

Fix integers $Q \geq 8$ and $3 \leq L < Q/4$, put $h = 4LQ$, and let $Q < q < 2Q$ be prime. The modeled support is

$$S_{q,L} = \left\{ mq^{-1} \pmod{h} : 0 < |m| \leq R_q,\ (m,h) = 1 \right\}, \qquad R_q = \left\lfloor \frac{Lq}{Q} \right\rfloor.$$

Since $R_q < 2L < q$, the support is internally injective. Write

$$A_h(R) = \#\{1 \leq m \leq R : (m,h) = 1\}, \qquad N_q = |S_{q,L}| = 2A_h(R_q).$$

For a common uniform atom amplitude, diagonal row mass is a common scalar times N_q. We call N_q the raw row mass, without identifying it with a physical source weight.

Lemma 1 (Universal comparability envelope)**.** *For every admissible clock,*

$$2 \leq N_q \leq 2(2L-1), \qquad 1 \leq \kappa_{\mathrm{raw}}(Q,L) := \frac{\max_{q \in \mathcal{P}_Q} N_q}{\min_{q \in \mathcal{P}_Q} N_q} \leq 2L-1,$$

whenever the prime shell $\mathcal{P}_Q$ is nonempty.

Proof. The shell inequalities give $L \leq R_q \leq 2L-1$. The multiplier 1 is always primitive, so $A_h(R_q) \geq 1$, while trivially $A_h(R_q) \leq R_q \leq 2L-1$. Taking the largest possible numerator and smallest possible denominator proves the claim. □

The lemma already shows that fixed comparability is not a formal consequence of the cutoff range: its best structure-only upper bound grows linearly in L. The next section constructs genuine active rows whose ratio diverges nearly linearly, up to one logarithm.

3 A critical primorial clock

Let

$$P_L = \prod_{\substack{\ell \leq L \\ \ell \text{ prime}}} \ell, \qquad j_L = \left\lfloor \frac{L \log L - \log P_L}{\log 2} \right\rfloor, \qquad Q_L = 2^{j_L} P_L. \tag{1}$$

The prime number theorem in the form $\log P_L \sim L$ makes j_L nonnegative for all large L. The floor in (1) gives

$$\log Q_L = L \log L + O(1), \qquad L \sim \frac{\log Q_L}{\log\log Q_L}. \tag{2}$$

Every prime divisor of Q_L is at most L.

Lemma 2 (Endpoint shell rows)**.** *For all sufficiently large L, there are primes p_L, r_L satisfying*

$$Q_L < p_L < Q_L + \frac{Q_L}{2L}, \qquad 2Q_L - \frac{Q_L}{2L} < r_L < 2Q_L. \tag{3}$$

Consequently $R_{p_L} = L$ and $R_{r_L} = 2L-1$.

Proof. The classical zero-free-region remainder gives, for some $c > 0$,

$$\pi(x) = \mathrm{Li}(x) + O\Big(xe^{-c\sqrt{\log x}}\Big)\,;$$

see, for example, Davenport [1], Montgomery and Vaughan [3] and Iwaniec and Kowalski [2]. Set $x = Q_L$ and $y = Q_L/(2L)$. The main term in $\pi(x+y) - \pi(x)$ is asymptotic to $y/\log x$, whereas the error-to-main ratio is

$$O\Big(L \log Q_L\, e^{-c\sqrt{\log Q_L}}\Big) = o(1)$$

by (2). This gives the first prime in (3); the interval immediately below $2Q_L$ is identical. Multiplying (3) by L/Q_L yields values strictly between L and $L+1/2$, and between $2L-1/2$ and $2L$, respectively. Taking floors proves the cutoff identities. □

Theorem 3 (Critical raw-mass separation)**.** *Along the clocks (1),*

$$N_{p_L} = 2, \qquad N_{r_L} = 2\{1 + \pi(2L-1) - \pi(L)\}, \tag{4}$$

and therefore

$$\kappa_{\mathrm{raw}}(Q_L, L) \geq 1 + \pi(2L-1) - \pi(L) \sim \frac{L}{\log L} \longrightarrow \infty. \tag{5}$$

Proof. Put $h_L = 4LQ_L$. Every $2 \leq m \leq L$ has a prime divisor at most L, hence a divisor of Q_L. Thus $A_{h_L}(L) = 1$.

If $L < m < 2L$ is composite, it has a prime divisor at most $\sqrt{m} < \sqrt{2L} < L$ for $L \geq 3$, so it is not primitive modulo h_L. If m is prime, then $m > L$ and it divides neither Q_L nor $4L$, so it is primitive. Hence the positive primitive multipliers up to $2L-1$ are exactly 1 and the primes in $(L, 2L)$. Lemma 2 now gives (4). The PNT on $(L, 2L)$ gives (5). □

Theorem 3 refutes a scale-independent raw comparability constant as a theorem of the modeled support. It does not refute comparability for a particular source after nonuniform weighting or normalization.

4 Finite reproduction and route consequence

The certificate evaluates four clocks with $L = 5, 7, 11, 13$. In each case $Q = 2^j P_L$ is chosen so that $\log Q$ is already close to $L \log L$. It locates the first prime in each endpoint interval, verifies the two cutoff identities, enumerates coprime multipliers, and checks

$$A_{4LQ}(L) = 1, \qquad A_{4LQ}(2L-1) = 1 + \pi(2L-1) - \pi(L).$$

The largest certified scale is

$$Q = 257955735797760, \qquad p = 257955735797867, \qquad r = 505990097141761.$$

The four exact raw mass ratios are $2, 3, 4, 4$. These small values are finite reproduction only; divergence is supplied by Theorem 3.

One implementation uses the deterministic unsigned-64-bit Miller–Rabin basis set and constructs the primorial clock. A separate implementation uses a different deterministic basis theorem and rebuilds every record from the serialized scale data. Normal and optimized Python modes agree. An adversarial script removes one required primorial factor from each clock and verifies that the low-row identity fails, while the cutoff and universal-envelope boundary checks remain exact. The record digest is

`93fec725ce67b017dffcf4b04ecb4125dd67b241d6e5cd12f18015fb0ee0f15c`.

The route implication is narrow but decisive. A collision-incidence estimate may be converted to an incident-mass estimate only after row weights are controlled. At critical depth, raw support count does not supply that control. The next minimal question is therefore whether unit-row normalization yields a uniformly stable collision operator, and only afterward whether such normalization is source-valid.

5 Conclusion

Critical resonance depth does not automatically produce a well-conditioned raw row family. Primorial saturation and endpoint prime rows yield an explicit sequence on which the support-mass comparability constant grows at least as $L/\log L$. Thus the fixed-comparability transfer is a genuine extra hypothesis rather than a consequence of clock geometry.

This obstruction points to a concrete repair instead of closing the route: normalize rows and analyze the resulting collision Gram operator. Such normalization changes the source interface and therefore cannot be silently imposed on the physical V59 coefficients. No collision lower bound, signed cancellation, L^2 estimate, strict $1/400$ payment, full Gate B, or twin-prime conclusion is asserted here.

References

[1] Harold Davenport. *Multiplicative Number Theory.* Springer, third edition, 2000.

[2] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications.* American Mathematical Society, 2004.

[3] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory.* Cambridge University Press, 2007.